

Operating Systems and Networking

PLD SMART

IDENTIFICATION

CODE : IF-4-PLD-SMART
ECTS : 4.0

HOURS

Lectures : 1.5 h
Seminars : 4.0 h
Laboratory : 32.0 h
Project : 0.0 h
Teacher-student
contact : 37.5 h
Personal work : 60.0 h
Total : 97.5 h

ASSESSMENT METHOD

Oral examination [45 minutes for each group of 6 students] + a 2 minutes video recording to present the project outcomes

TEACHING AIDS

- Open data provided by GrandLyon
- specific documents available on the moodle platform

TEACHING LANGUAGE

French

CONTACT

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AIMS

The project's main goal is to stimulate students' creativity by leading a complex project with societal impacts in the domain of smart cities. This project emphasises on specific invariants related to engineers in ICT:

- Strong technical expertise: students are confronted with a complex problem requiring technical skills acquired in the last two years of the programme. these includes multidisciplinary topics from different areas such as software development, data mining, image processing, project management, team work etc.
- capabilities to develop innovative solutions for the society: the complex problem requires the creation of a useful service to deliver to inhabitants in smart cities (residents, visitors or managers)
- Self-management: students design and develop a quality demonstrator and apply an agile project management process, a risk assessment and mitigation, a rigorous time management
- Communication: students establish an effective communication strategy within the team [mobilizing, exchange, use the skills of the other], and with external actors [instructors, external jury members, experts, ...] - generating client "empathy".

Acquired Skills;

- Design complex systems using problem and domain related languages and tools in the field of computer science.
- Develop distributed complex technical and software architectures techniques;
- Manage software development projects
- Analyse business and customer requirements
- Interact and collaborate with project team members
- Conduct brain storming sessions to develop creative and innovative solutions
- Meet managerial and technical goals in complex problem-solving problems

CONTENT

- A. A short course [1.5 hours] to kick off the project as a call for a tender "Development of an innovative digital service for a smart city (Grand Lyon)"
- B. A session [2hours /group of 30 students] to assist students to write a response to the tender with a particular focus on the following aspects: innovation, business model, value analysis, market research
- C. Individualised tutoring [1 hour per group consisting of 6 students] as support for point B
- D. 8 "technical" sessions reserved to the effective implementation of the project and to deliver a demonstrator

BIBLIOGRAPHY

<http://data.grandlyon.com/>

The bibliography depends on the digital service to be developed and its technical requirements